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Regression function

For square loss function $L(Y, f(X)) = (Y - f(X))^2$, the expected prediction error is

$$\text{EPE}(f) = \mathbb{E}[(Y - f(X))^2].$$

It is easy to see that the regression function $f(x) = \mathbb{E}[Y \mid X = x]$ minimizes the expected prediction error. Indeed, as all $\sigma(X)$ -measurable random variables are of the form $f(X)$, it is equivalent to minimize $\|Y - Z\|_2$ over all $\sigma(X)$ -measurable random variables Z . This is of course achieved by the orthogonal projection of Y onto the closed subspace of $\sigma(X)$ -measurable functions which is exactly $\mathbb{E}[Y \mid X]$.

The nearest neighbour method directly tries to estimate this regression function. In particular, intuitively, given iid $(X_i, Y_i)_{i=1}^n$ and x , we hope to approximate

$$\mathbb{E}[Y \mid X = x] \approx \frac{1}{k} \sum_{i=1}^n \mathbf{1}_{\{\|X_i - x\| \leq \|X_{(k)} - x\|\}} Y_i.$$

Namely, we take the sample average of the Y_i values corresponding to the k nearest neighbours of x . By taking a variance-bias decomposition, if we take k, n in such a way that $k \rightarrow \infty$ and $\frac{k}{n} \rightarrow 0$, then under some mild assumptions on the distribution of (X, Y) , we have the approximation converges in the limit.

The nearest neighbour method (and local methods in general) is problematic in high dimensions. For fixed x , if we want our estimate to depend on the data within a neighbourhood of x which captures a fixed proportion of the data, then the radius of this neighbourhood has to grow exponentially in the dimension (so rather contradicts the intuition of locality). This is called *the curse of dimensionality*.

A natural extension of the nearest neighbour method is, instead of taking the average of the nearest neighbours, we can take a weighted average of all points weighted by their distance to x . We call the weight function K the kernel function and a simple estimate (Nadaraya-Watson) is

$$f(x) = \frac{1}{Z} \sum_{i=1}^n K(x, X_i) Y_i$$

with the normalisation $Z = \sum_{i=1}^n K(x, X_i)$.

Linear regression

Let us consider the OLS estimator from a probabilistic perspective. Suppose $\mathcal{H}, \mathcal{K}$ are two Hilbert spaces and we have random variables $X \in \mathcal{H}$ and $Y \in \mathcal{K}$. The fundamental assumption one makes with linear regression is that the regression function is linear, i.e. there exists a linear operator $\beta : \mathcal{H} \rightarrow \mathcal{K}$ such that $\mathbb{E}[Y | X] = \beta X$ (we assume w.l.o.g. that both X, Y are centred).

Now, recalling that the covariance operators C_X and C_{XY} are defined by

$$C_X : \mathcal{H} \rightarrow \mathcal{H}, \quad C_X f = \mathbb{E}[\langle f, X \rangle_{\mathcal{H}} X]$$

and

$$C_{XY} : \mathcal{H} \rightarrow \mathcal{K}, \quad C_{XY} f = \mathbb{E}[\langle f, X \rangle_{\mathcal{H}} Y].$$

Thus, we compute

$$\begin{aligned} \langle C_{XY} h, k \rangle &= \mathbf{E}[\langle X, h \rangle \langle Y, k \rangle] = \mathbf{E}[\langle X, h \rangle \langle \mathbf{E}[Y | X], k \rangle] \\ &= \mathbf{E}[\langle X, h \rangle \langle \beta X, k \rangle] = \langle \beta C_X h, k \rangle \end{aligned}$$

for all $h \in \mathcal{H}$ and $k \in \mathcal{K}$. Hence, we have $C_{XY} = \beta C_X$. Thus, in the case where C_X is invertible, we have $\beta = C_{XY} C_X^{-1}$.

In the finite-dimensional case, the OLS estimator is exactly the empirical version of this formula. In particular, suppose now $\mathcal{H} = \mathbb{R}^p$ and $\mathcal{K} = \mathbb{R}$. Then, we have the covariance matrices $C_X = \mathbb{E}[XX^\top]$ and $C_{XY} = \mathbb{E}[XY^\top]$. Now, given iid data $(\mathbf{X}, \mathbf{Y}) = (X_i, Y_i)_{i=1}^n \in \mathbb{R}^{n \times p} \times \mathbb{R}^n$, we can estimate C_X and C_{XY} by their empirical versions

$$\hat{C}_X = \frac{1}{n} \sum_{i=1}^n X_i X_i^\top = \frac{1}{n} \mathbf{X}^\top \mathbf{X}, \quad \hat{C}_{XY} = \frac{1}{n} \sum_{i=1}^n X_i Y_i^\top = \frac{1}{n} \mathbf{X}^\top \mathbf{Y}.$$

Thus, the OLS estimator is exactly $\hat{\beta} = \hat{C}_{XY} \hat{C}_X^{-1}$. Note that the OLS estimator one typically writes down is the transpose of what we have here as one typically instead assumes $Y = X\beta + \varepsilon$.

The Cameron-Martin space and the RKHS

Suppose now X is a Gaussian random variable on some Banach space $\mathcal{B}$ with law μ . We recall that the Cameron-Martin space $\mathcal{H}_\mu$ of μ is defined as the closure (with respect to the Cameron-Martin norm) of the pre-Cameron-Martin space

$$\mathcal{H}_\mu^\circ = \{x \in \mathcal{B} : \exists x^* \in \mathcal{B}^*, l(x) = C_\mu(l, x^*) \forall l \in \mathcal{B}^*\} = C_\mu(\mathcal{B}^*)$$

where in the latter equality we identify $C_\mu : \mathcal{B}^* \rightarrow \mathcal{B}$ by $C_\mu l = \int l(x) x \mu(dx)$ with the integral understood in the Bochner sense.

We show that the Cameron-Martin space is exactly the RKHS (in the statistician's sense) associated to the kernel $k(h, k) = C_\mu(h, k) = \int h(x) k(x) \mu(dx)$ (to be more precise, the RKHS of k is the canonical identification of $\mathcal{H}_\mu$ in $\mathcal{B}^{**}$). Note that the RKHS here is different from the RKHS in [Haia, Haib] which is instead the $L^2(\mu)$ -closure of $\mathcal{B}^*$.

Recall that the RKHS $\mathcal{H}$ associated to a reproducing kernel $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is a Hilbert space of functions $f : \mathcal{X} \rightarrow \mathbb{R}$ such that $k(x, \cdot) =: k_x \in \mathcal{H}$ for all x and

$$\langle f, k(x, \cdot) \rangle_{\mathcal{H}} = f(x).$$

In general, the RKHS $\mathcal{H}$ is canonically defined by $\mathcal{H} = \overline{\mathcal{H}^\circ} = \overline{\langle k(x, \cdot) : x \in \mathcal{X} \rangle}^{\mathcal{H}}$ with the closure taken with respect to the topology generated by the inner product

$$\left\langle \sum_i \alpha_i k(x_i, \cdot), \sum_j \beta_j k(x_j, \cdot) \right\rangle_{\mathcal{H}} = \sum_{i,j} \alpha_i \beta_j k(x_i, x_j).$$

In this case, $\mathcal{X} = \mathcal{B}^*$ and $\mathcal{H} = \mathcal{H}_\mu$ with the canonical identification $x \in \mathcal{H}_\mu \subseteq \mathcal{B}^{**}$, $x : l \mapsto l(x)$. Indeed, in this case, $k(l, \cdot) = C_\mu l \in \mathcal{H}_\mu^\circ$ and for any $x \in \mathcal{H}_\mu^\circ$, we have that

$$\langle x, k(l, \cdot) \rangle_\mu = \langle x, C_\mu l \rangle_\mu = C_\mu(x^*, l) = l(x) = x(l)$$

where we used the fact that $(C_\mu l)^* = l$.

Conversely, suppose we have the reproducing kernel $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ (recall it is symmetric and positive definite) and its associated RKHS $\mathcal{H}$. In this case, we consider a isonormal Gaussian process W on $\mathcal{H}$ (and its first Wiener chaos $\mathcal{H}_1$ is isomorphic to $\mathcal{H}$ via the map $h \mapsto W(h)$). On the other hand, under some mild regularity assumptions (e.g. via Kolmogorov continuity criterion), there exists a Gaussian measure μ on some Banach space $\mathcal{B} \subseteq \mathbb{R}^{\mathcal{H}}$ for which the Cameron-Martin space is exactly $\mathcal{H}_1$. Thus, in this case, the above correspondence between the Cameron-Martin space and the RKHS holds as well.

Ridge regression and kernels

The OLS estimator is the unbiased estimator minimising MSE. However, it is not the estimator minimising MSE among all estimators. In particular, via the bias-variance decomposition, one might hope to reduce the MSE by introducing some bias. The ridge regression estimator achieves this by penalising large values of β . This is intuitive as β is large when C_X is close to singular and thus the OLS estimator is unstable. The ridge regression estimator instead optimises

$$\hat{\beta} = \arg \min_{\beta} \left(\frac{1}{n} \sum_{i=1}^n (Y_i - X_i^\top \beta)^2 + \lambda \|\beta\|^2 \right)$$

for some regularisation parameter $\lambda > 0$.

References

Hairer:Malliavin

[Haia] Martin Hairer. Advanced stochastic analysis. Lecture notes.

Hairer:SPDEs

[Haib] Martin Hairer. An introduction to stochastic PDEs. Lecture notes.